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POSTNATAL EXERCISE INTERVENTIONS: A SYSTEMATIC REVIEW OF ADHERENCE AND EFFECT

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POSTNATAL EXERCISE INTERVENTIONS: A SYSTEMATIC REVIEW OF ADHERENCE AND EFFECT

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ABSTRACT

Objectives To evaluate adherence to and effect of postnatal physical activity (PA) interventions

Setting Systematic review of PA intervention RCTs in postnatal women

Participants The 24 identified studies included 1531 postnatal women in community and secondary care settings. Studies were included if the PA interventions was commenced and assessed in the year after delivery of a baby. Where the PA intervention was restricted to pelvic floor exercises, physiotherapy or single exercise sessions, studies were excluded.

Outcomes measures Primary outcome: adherence to the PA intervention. Secondary outcomes: effect of PA interventions on the studies' specified primary outcome. Subgroup analysis was undertaken for supervised and unsupervised exercise interventions

Results 1404 records were screened for potential study inclusion, full-text review was performed on 140 articles. Of these, 24 studies were included. Studies were generally small, median N=66 (20-130). PA interventions used were highly varied, targets for PA per week ranged from 60 to 275 minutes per week. When reported, for the groups undergoing PA interventions, mean adherence was 80.4% and mean lost-to-follow-up (LTFU) was 11.8%; 6/9 unsupervised intervention studies and 5/15 supervised intervention studies showed a significant effect of the PA intervention tested on their specified primary outcome. LTFU was higher (14.5% vs 10%) and adherence to the intervention was lower (73.6% vs. 86%) for unsupervised vs. supervised studies.

Conclusions Studies of PA interventions inconsistently reported adherence and LTFU. Where multiple studies evaluate PA interventions on an outcome, they had inconsistent effects, with generally low study quality and high risk of bias. Supervised studies appeared less effective than unsupervised studies, perhaps reflecting reduced self-efficacy. Reassuringly, studies reported no adverse effect of PA interventions. Further, high-quality research reporting adherence and LTFU is needed into how and when to deliver postnatal PA interventions to benefit postnatal physical and mental health.

Registration number PROSPERO CRD42019114836

Strengths and Limitations

- This review considered only PA intervention studies which were compared to a control group with no PA intervention or with advice to be sedentary, to allow evaluation of effect of the PA intervention on the primary outcomes of each study.
- There may be studies which compare PA interventions in groups other than postnatal women which had high adherence and positive or negative effects and will have been excluded from our analysis.
- We considered only PA interventions in the year after delivery, potentially excluding interventions later after delivery which may be appealing to women and effective in improving physical and mental health.

Take home messages

- Studies using physical activity interventions need to report LTFU and adherence.
- Further research is needed into how/when to deliver postnatal PA interventions for benefit on physical and mental postnatal health.
- Studies reported no adverse effect of PA interventions on women or their children
- PPI and stakeholder involvement was lacking in the identified studies
- Studies must report using internationally agreed standards and outcome sets to improve their utility to women and their healthcare professionals

INTRODUCTION

Physical inactivity is estimated to cause between 6 and 10 percent (%) of the major non-communicable diseases of breast and colon cancer, type 2 diabetes, and coronary heart disease; and 9% of premature

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mortality worldwide (1). Not surprisingly, physical inactivity is responsible for 1 in 6 UK deaths and is estimated to cost the UK £7.4 billion annually (2). This is exemplified in UK women where 42% are currently not active enough to maintain good health (2).

Adult physical activity (PA) guidelines recommend 150 minutes of moderate-to-vigorous intensity physical activity (MVPA) per week; only 67% of UK women aged 35-44 achieve this with further reductions with age (3). Disparity also exists from a young age; the percentage of girls aged 11 achieving the recommended 60 minutes of daily MVPA declines from 20% to 12% by the age of 15; in boys this falls from 32% to 25% (4).

Women's physical activity levels are low during and after pregnancy. Sixty percent of women during pregnancy report engaging in no leisure time PA (5). Fewer than 13% of women meet recommended PA levels during pregnancy, and this decreases with each trimester (6). PA measured during and after pregnancy is predominately of a low intensity , providing somewhat limited health benefits (7). It is important to note that postnatal total PA (measured in hours or weeks) is similar to that seen in mid pregnancy, but comprises a larger proportion of care-giving activities (7).

The postnatal time is a difficult period for women to restart or maintain PA. Fatigue from poor sleep and the demands of infant feeding can also contribute to lower PA levels. The Institute of Health Visitors advise waiting for the 6-week postnatal visit with women's general practitioners (GPs) before resuming high-impact exercise. GPs have to assess maternal and child health at this visit, though this visit is generally short and advice given can be extremely variable (8,9). Healthcare professionals need to know which interventions they can best recommend to post-partum women, enabling them to meet recommended levels of PA and optimise their health.

The UK Chief Medical Officers recommends that pregnant and postnatal women undertake 150 minutes moderate PA per week, and have provided useful infographics which are available as a guide for women and healthcare professionals (10,11). These are endorsed by the American College of Obstetricians and Gynaecologists' (ACOG) guidelines, who also recommend 20 to 30 minutes of PA per day during pregnancy on most or all days of the week (12), and briefly describe the postpartum period as an opportune time to gradually resume an exercise routine. Clearly however, the figures above indicate that this is not being achieved.

To determine the most effective physical activity intervention, we conducted a systematic review to determine adherence to postnatal physical activity interventions, and the effect of the interventions on the outcomes investigated in the studies identified. The results would also inform healthcare professionals who advise women on the most efficacious PA interventions for desired effect.

We hypothesised that adherence to PA interventions would be greater for supervised (in person supervision or with monitoring of PA e.g. a pedometer) vs. unsupervised interventions; and that the effects of PA interventions on the primary outcomes of studies would be related to adherence.

METHODS We carried out a systematic review based on a pre-published protocol (PROSPERO CRD42019114836) and reported the findings in accordance with specifications recommended by the Meta-Analysis of Observational Studies in Epidemiology (MOOSE) guidelines (13).

Search strategy and selection criteria Embase, MEDLINE, and Cochrane Central Register of Controlled Trials (CENTRAL) databases were searched in duplicate as per the search strategies (supplementary files 3,4 and 5), publication date was not restricted. The references of included articles were also hand-searched by the reviewers to highlight further studies for potential inclusion.

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An initial screen of titles and abstracts was performed independently by two reviewers (SS and EM), to identify articles for further review. Articles were included after full text review if they were randomised, controlled trials which compared moderate and/or vigorous PA to no PA. This could be in combination with another intervention e.g. a dietary intervention, if a comparison group with the same dietary intervention alone was tested. If more than one intervention was included, it was essential that there was a comparison between exercise and no exercise groups. If there was any ambiguity regarding a study's inclusion, a third reviewer (AM) decided on eligibility. Where data was missing or incomplete, contact with the corresponding author was made by email.

Where the PA intervention was restricted to pelvic floor exercises, physiotherapy or single exercise sessions, studies were excluded. Studies were excluded from the review if they did not report on original data or were animal studies. Studies were not excluded based on language, study type or publication status. There was no other restriction on the type of PA intervention.

Data analysis Two reviewers (SS and EM) extracted the data from the included studies using a pre-specified extraction template. Extracted data elements recorded for each study included general information, eligibility criteria, study characteristics, (setting, country, year, design, type and duration of PA intervention, sample size), adherence to the intervention, and primary outcome measures.

Both reviewers assessed risk of bias for each study independently using the Cochrane Collaboration's tool for assessing risk of bias (14). The following domains were assessed as having a high, low or unclear risk of bias: random sequence generation, allocation concealment, blinding of participants/ personnel, blinding of outcome, incomplete outcome, selective reporting and overall bias. The reviewers then used the GRADE criteria (13) to assess the quality of the evidence within the review for the specific outcomes identified.

RESULTS

After the initial search, 1404 records were screened for potential study inclusion. After title and abstract review, full-text review was performed on 140 articles. Of these, 24 articles reporting the effect of randomised controlled trials of postnatal exercise interventions vs. no exercise were included (Supplementary File 2, Figure1) . Study designs included; control vs. exercise; control vs. dietary intervention vs. dietary intervention and exercise; and control vs. dietary intervention vs. exercise intervention vs. dietary and exercise intervention (Supplementary File 1, Table 1). The 24 studies included 1531 postnatal women.

Studies included were conducted in North America, Europe, Africa, Asia and Australasia. Studies were generally small, ranging from 20 to 130 participants, median *N*=66. Physical Activity interventions used were varied, including provision of active video games (WiiFit and balance board, (15)), exercise prescriptions (16–20), CDs and DVDs (21–23), telephone (24,25) and face-to-face counselling sessions (25–29), group classes (22,23,30,31) and supervised exercise sessions (31–33). Interventions varied in duration between 11 days and 10 months. Where reported, the targets for PA per week ranged from 60 to 275 minutes per week. One study prescribed exercise to attain an individualised energy deficit. Four studies did not report a PA target. A meta-analysis was not performed due to the heterogeneity of the studies identified.

Of the studies included, 9 used unsupervised exercise interventions (15,19,20,22–25,29,30), and 15 used exercise with some form of supervision. 6 studies used PA monitoring (a pedometer or an Actigraph, (16,17,27,28,34,35)), 1 used PA monitoring and in-person supervised exercise interventions (18), 6 used in-person supervised interventions (31–33,36–38)and 2 used a mixture of in-person supervised and unsupervised interventions (21,26). (One study used an actigraph for 7 days prior to follow-up 6 month post intervention though not during the intervention (21).)

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Loss to follow-up (LTFU)in the exercise intervention group was reported in 17/25 studies (16,17,19–21,24–31,33–35)and adherence to the PA intervention was reported by 11/25 studies (16,18,19,25,27,29–31,35,38). When reported, for the groups undergoing PA interventions, mean LTFU was 11.8% and mean adherence was 80.4%. Some studies reported only overall rates of LTFU and adherence. In studies where LTFU and adherence is reported, the proportions were similar for intervention and control/comparison groups.

Effect of PA intervention on weight loss was reported by 2/6 studies (15,30), with weight loss between 2.1 and 5.6 kg reported (Supplementary File 1, Table 2). In studies which compared dietary, PA and combined interventions, PA interventions were not associated with weight loss over and above the effect of the dietary intervention(16,34). Higher rates of LTFU and lower rates of adherence (where reported) were seen in studies which showed effect on weight loss.

For measures of postnatal depression, 3/7 reported an effect of a PA intervention (18,21,25), the effect size varied greatly, from -4.79 to -12.8 points on the Edinburgh Postnatal depression scale. 2 out of 3 of these studies reported LTFU, 8% in both, adherence was reported by 2/3 and was 64 and 87%.

Physical fitness, defined by VO_2max and investigated by 2 small studies, is improved by PA interventions (36,37). General fatigue, part of the Multidimensional Fatigue Inventory, appears to be reduced by PA interventions (17,23). Lactation is not impaired by PA interventions in studies with 90 and 95% adherence (35,38), nor is infant growth (33). Glycaemia (20) and sleep quality (22)appear to be improved by PA interventions. Bone density (19)and (surprisingly) physical activity (29) were not significantly affected by PA interventions, although the latter study had n=29 and an adherence of 40% to the intervention.

Risk of bias assessment found that most studies were at high (7/24) or unclear (16/24) risk of bias (Supplementary File 1, Table 3). Blinding measures for personnel and participants were absent for all studies, conceding that it would be difficult to do so, especially for supervised exercise interventions. No studies referred to a published protocol and/or was convincing that the published report included all expected outcomes, including those that were pre-specified. Many studies did not list a primary outcome, in these cases we used the title or outcome reported first or with most detail in the results section.

Study quality for the various outcomes tested was generally low or very low by GRADE score (Table 3), meaning we had low or very low confidence in the effect estimates. We had moderate confidence in the effect estimates for PA interventions having no effect on lactation outcomes (milk volume produced and milk volume consumed).

Sub-group analysis of unsupervised and supervised studies 6/9 unsupervised intervention studies (15,20,22,23,25,30) and 5/15 supervised intervention studies (17,18,26,36,37) showed an effect of the PA intervention tested on their specified primary outcome (Supplementary File 1, Table 4.) LTFU was higher (14.5% vs 10%) and adherence to the intervention was lower (73.6% vs. 86%) for unsupervised vs. supervised studies.

Discussion In this systematic review, we show that PA interventions have inconsistent effects on most outcomes, with generally low study quality and high risk of bias. Adherence was reported by only 9/24 studies, and LTFU for 16/24 studies. For any intervention, the LTFU and adherence must be reported to allow judgement on whether the intervention was acceptable to participants and whether there can be confidence in the effect, or lack of it, shown. Adjustment for known confounders, such as pre-existing exercise habits, physical fitness, weight, BMI were either not present or unclear in the majority of studies. This could have contributed to the heterogeneity of study conclusions.

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Supervised exercise interventions did not appear to be more effective on the various outcomes tested than unsupervised interventions, however the studies testing them had lower LTFU and higher adherence, which increases our confidence in the outcomes of the studies. This was unexpected and may reflect that women may feel more empowered when allowed to exercise on their own terms; self-efficacy may lead to greater efficacy of the intervention.

Women considering postnatal physical activity can be counselled that physical activity is unlikely to impact on breastfeeding or infant growth. Additionally, no studies led to any serious adverse events. Whilst the wide range of interventions makes recommendations on the types of exercise difficult to assess, it also demonstrates the inventiveness of researchers in the field who have considered using video games (15), pilates (22,23), tai chi (20) and telephone-based interventions (24) in addition to more conventional exercise counselling and supervised exercise classes.

Since the majority of these studies were conducted, there are now many wearable devices available, which could improve objective assessment of PA in intervention studies. These will allow accurate assessment of adherence and measures of fitness to be measured, such as morning resting heart rate. Our group is conducting feasibility studies before, during and after pregnancy using wearables and phone applications to monitor cardiovascular health.

Further research is needed into the optimal timing, method of delivery and content of postnatal PA interventions to maximise recruitment, adherence and ensure effects on outcomes are effectively evaluated. Pregnancy is a risk factor for weight gain and complications of pregnancy, such as pre-eclampsia and gestational diabetes, reveal groups of women likely to benefit their long-term health from PA interventions.

More generally, the postnatal period is a difficult time restart and maintain physical activity. There are likely to be population benefits to short and long term risk of non-communicable disease from public health approaches which encourage and enable women to increase their PA after having a baby, however studies to date have not consistently demonstrated these. None of the included studies referred to Public and Patient Involvement (PPI) in design or conduct of their research. PPI is invaluable to ensure recruitment, adherence and rigorous evaluation with outcomes which are relevant to women and their healthcare providers.

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Competing Interests

The authors declare that that they have read and understood the BMJ Group policy on declarations of interest and have no competing interests to declare.

Contributorship statement

EM conceived the study. EM, AM and SS designed the study. EM and SS undertook the literature search, review and data extraction. EM, AM and SS analysed the data. EM and SS wrote the first draft. EM, AM and SS edited and wrote the final draft.

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PPI

PPI for a separate study helped to conceive the research question for this study. We did not undertake PPI for this study.

Dissemination and data sharing

We will disseminate the information in this study to women in our area via our institutional news and social media channels. We will share our data on request to the corresponding author, EM.

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	Outcome category	Author	Setting	Country	N	Design	Intervention group 1	Intervention group 2	Duration intervention	Supervised/ monitored/ unsupervised	PA target per week	Comparison group 1	Comparison group 2	Primary outcome measure
1	**Weight, Body composition**	Brekke et al. (2014) *(16)*	Community	Sweden	*68*	C, D, E or DE.	45 min brisk walk 60-70% MHR*4/week monitored by HR monitor and diary.	as for group 1 +dietary intervention	6 months	HR monitor	180 mins	Usual care	Dietary intervention	Waist circumference (cm)
		Kinnunen et al. (2007) *(26)*	Community	Finland	*85*	C, E	Counselling sessions *5 at 2, 3, 5 6 and 10 months postnatal. Option of attending weekly supervised exercise class.		8 months	Unsupervised, optional supervised	150 mins	Usual care		% return to normal weight at 10 months postnatal
		Tripette et al. (2014) *(15)*	Community	Japan	*34*	C, E	Active video games for 30 mins day (WiiFit and balance board)		40 days	Unsupervised	Not reported	Usual care		Weight loss kg
		O'Toole et al. (2003) *(30)*	Community	USA	*40*	D, DE	Group educational sessions *17-23 on PA and diet, combined effect 500kcal daily energy deficit		6-10 months	Unsupervised	Energy deficit of 1050kcal	Dietary counselling session *1, 500kcal/ day energy defecit		Weight loss kg
		Bertz et al. (2012) *(34)*	Community	Sweden	*68*	C, D, E or DE.	E 45 min brisk walk 60-70% MHR*4/week monitored by HR monitor and diary	as for group 1 +dietary intervention	12 weeks	HR monitor	180 mins	Usual care	Dietary intervention 500 kcal/day deficit	Weight loss kg
		Osman et al. (2016) *(32)*	Community	Egypt	*47*	D, DE	Supervised aerobic exercise sessions *3/week for 4 weeks to 60-70% MHR, dietary and BF advice		4 weeks	Supervised	105 mins	Dietary and BF advice		Weight loss kg
2	**Postnatal depression**	Da Costa et al. (2009) *(27)*	Community	Canada	*88*	C, E	Counselling sessions *4 with exercise physiologist over 12 weeks. 60-120mins/week at 60-85% MHR		12 weeks	HR monitor	120 mins	Usual care		EPDS
		Heh et al. (2008) *(21)*	Community	Taiwan	*80*	C, E	Exercise program CD for 2 home exercise sessions/ week, 1 hour weekly exercise session at hospital		3 months	Supervised and unsupervised	180 mins	Usual care		EPDS
		Lewis et al. (2014) *(24)*	Community	USA	*130*	C, E	11 telephone sessions to help women increase MVPA,		6 months	Unsupervised, Actigraph 7 days	Not reported	Usual care		% depression by SCID-1
		Daley et al. (2015) *(25)*	Community	England	*94*	C, E	2* exercise counselling sessions, 2* phone calls, leaflets months 3,4,5 and 6		6 months	Unsupervised	150 mins	Usual care		EPDS
		Daley et al. (2008) *(28)*	Primary care, community	England	*38*	C, E	2* exercise counselling sessions, 2* phone calls		12 weeks	Pedometer	105 mins	Usual care		EPDS
		Armstrong et al. (2003) *(18)*	Community	Australia	*20*	C, E	Group Pram walking 3*/week, 30–40 mins. at 60–75% of MHR, weekly social support session		12 weeks	Supervised, Pedometer	90-120 mins	Usual care		EPDS
		Norman et al. (2010) *(31)*	Community	Australia	*161*	C, E	1*exercise class/week. 1* General health advice education session		4 weeks	Supervised	60 mins	1* General health education session		Positive affect balance scale

Outcome category	Author	Setting	Country	N	Design	Intervention group 1	Intervention group 2	Duration intervention	Supervised/ monitored/ unsupervised	PA target per week	Comparison group 1	Comparison group 2	Primary outcome measure
3 Physical fitness	Zourladani et al. (2012) *(37)*	Home	Greece	*37*	C, E	3* exercise classes/week, 50-60mins aerobic and strengthening exercise		12 weeks	Supervised	150-180 mins	Usual care		VO2max (ml/kg/min)
	Lovelady et al. (1995) *(36)*	Community	USA	*33*	C, E	5* exercise sessions/week, 45 mins aerobic exercise sessions		12 weeks	Supervised	225mins	Usual care		VO2max (ml/kg/min)
4 Fatigue	Dritsa et al. (2008) *(17)*	Community	Canada	*88*	C, E	4*Counselling sessisons to produce exercise prescription.		12 weeks	HR monitor	120mins	Usual care		Multidimensional Fatigue Inventory (MFI-20) -general fatigue
	Ashrafinia et al. (2015) *(22)*	Home	Iran	*80*	C, E	4 antenatal pilates training sessions, 5*home pilates sessions/week, 30 mins, exercise leaflet and CD		8 weeks	Unsupervised	150 mins			(MFI-20) -general fatigue
5 Lactation	Dewey et al. (1994) *(38)*	Community	USA	*33*	C, E	5* exercise sessions/week, individually tailored, gradated increase to 45 mins,		12 weeks	Supervised	225mins	Usual care		Infant milk intake g/kg/day
	McCrory et al. (1999) *(35)*	Community	USA	*67*	C, D, or DE.	Exercise 9/11 days at 50-70% maximum HR	as for group 1 +dietary intervention	11 days	HR monitor	Not reported	Usual care	Dietary intervention	Milk volume
6 Glycaemia	Youngwanichsetha et al. (2013) *(20)*	Hospital	Thailand	*64*	C, E	3*tai chi qigong sessions/week, 50mins		12 weeks	Unsupervised	150 mins	Usual care		Fasting plasma glucose (mg/dl)
7 Physical activity	McIntyre et al. (2012) *(29)*	Community	Australia	*28*	C, E	1*face to face exercise consultation, 8 telephone calls		12 weeks	Unsupervised	150 mins	Usual care		Change in PAmins/week (median)
8 Sleep quality	Ashrafini et al. (2015) *(22)*	Home	Iran	*80*	C, E	4 antenatal pilates training sessions, 5*home pilates sessions/week, 30 mins, exercise leaflet and CD		8 weeks	Unsupervised	150 mins	1*postnatal care advice session		Pittsburg Sleep Index (global score)
10 Infant growth	Lovelady et al. (2000) *(33)*	Community	USA	*48*	D, DE	4*exercise sessions/week, 45 mins, at 65-80% MHR		10 weeks	Supervised	180 mins	Usual care		Gain in Infant Weight (g)
11 Bone density	Lovelady et al. (2009) *(19)*	Home	USA	*20*	C, E	6*exercise sessions/week, 45 mins, 65-80% MHR, aerobic and resistance		16 weeks	Unsupervised	270 mins	Usual care		Change in whole body bone mineral density (g/cm2)

Key: C-control, E-Exercise intervention, D-dietary intervention, DE-ietary and exercise intervention, HR-heart rate, MHR-maxiumum heart rate, CV-cardiovascular, MVPA moderate or vigorous physical activity, PA -physical activity, FU-follow-up, BF-breast feeding

Table 1. Study characteristics, interventions, comparison groups and outcomes

Outcome category	Author	Primary outcome measure	LTFU intervention group(s)	Adherence to intervention	Effect in Intervention group 1	Effect in Intervention group 2	Comparison group 1	Comparison group 2	Exercise effect on primary outcome? (by CI or p<0.05)
1 Weight, Body composition	Brekke et al. (2014) *(16)*	Waist circumference (cm)	1/34 (3%)	62/68 (91%)	-4.3	-8.8	-1.3	-8.7	**No**
	Kinnunen et al. (2007) *(26)*	% return to normal weight at 10 months postnatal	5/53 (9%)	Not reported	50		30		**No**
	Tripette et al. (2014) *(15)*	Weight loss kg	Not reported	Not reported	-2.2		-0.5		**Yes (p<0.001)**
	O'Toole et al. (2003) *(30)*	Weight loss kg	4/21 (19%)	17/21 (81%)	-5.6		-0.7		**Yes (p<0.05)**
	Bertz et al. (2012) *(34)*	Weight loss kg	2/34 (6%)	Not reported	-2.4	-6.9	-0.8	-8.3	**No**
	Osman et al. (2016) *(32)*	Weight loss kg	Not reported	Not reported	-1.56		+0.09		**No**
2 Postnatal depression	Da Costa et al. (2009) *(27)*	Ediburgh Postnatal Depression Scale (EPDS)	6/46 (13%)	35/46 (76%)	-4.6		-5		**No**
	Heh et al. (2008) *(21)*	EPDS	2/35 (5.7%)	Not reported	-6.4		-3.5		**Yes (p=0.01)**
	Lewis et al. (2014) *(24)*	% depression by SCID-1	5/66 (8%)	Not reported	8.2		7.94		**No**
	Daley et al. (2015) *(25)*	EPDS	4/47(8.5%)	41/47 (87%)	-4.79		-2.83		**Yes (p=0.03)**
	Daley et al. (2008) *(28)*	EPDS	4/20 (20%)	Not reported	-4.9		-4.6		**No**
	Armstrong et al. (2003) *(18)*	EPDS	Not reported	64% attendance	-12.8		-3.7		**Yes (p<0.01)**
	Norman et al. (2010) *(31)*	Positive affect balance scale	2/62 (3%)	130/130 (100%)	+1.19		-0.18		**No**
3 Physical fitness	Zourladani et al. (2012) *(37)*	VO2max (ml/kg/min)	Not reported	Not reported	not reported		not reported		**Yes (p<0.001)**
	Lovelady et al. (1995) *(36)*	VO2max (ml/kg/min)	Not reported	Not reported	+6.8		+1.3		**Yes (p<0.0001)**
4 Fatigue	Dritsa et al. (2008) *(17)*	MFI-20 -general fatigue	6/46 (13%)	Not reported	-3.87		-1.83		**Yes (p0.016)**
	Ashrafinia et al. (2015) *(22)*	MFI-20 -general fatigue	Not reported	Not reported	-3.35		+1.2		**Yes (p<0.001)**
5 Lactation	Dewey et al. (1994) *(38)*	Infant milk intake g/kg/day	Not reported	149/165 sessions completed (90%)	117		118		**No**
	McCrory et al. (1999) *(35)*	Milk volume g/day	2/44 (5%)	21/22 (95%)	-16		+17	-1	**No**
6 Glycaemia	Youngwanichsetha et al. (2013) *(20)*	Fasting plasma glucose (mg/dl)	2/32 (6%)	Not reported	-10.25		-2.09		**Yes (p=0.02)**
7 Physical activity	McIntyre et al. (2012) *(29)*	Change in PA mins/week (median)	1/15 (7%)	6/14 (40%)	+60		+0		**No**
8 Sleep quality	Ashrafini et al. (2015) *(22)*	Pittsburg Sleep Index (global score)	Not reported	Not reported	5.45		8.35		**Yes (p<0.001)**
10 Infant growth	Lovelady et al. (2000) *(33)*	Gain in Infant Weight (g)	6/27 (22%)	Not reported	1925		1861		**No**
11 Bone density	Lovelady et al. (2009) *(19)*	Change in whole body bone mineral density (g/cm2)	3/10 (30%)	83.4% aerobic sessions	-0.8		-0.6		**No**

Key: EPDS - Edinburgh Postnatal Depression Scale, PA- Physical activity, MFI-20- Multidimensional Fatigue Inventory

Table 2. Study outcomes, Lost to follow-up, Adherence to interventions, effect of physical activity intervention studies on outcomes and adherence

	Outcome category	Study	Random sequence generation	Allocation concealment	Blinding participants/ personnel	Blinding outcome	Incomplete outcome ascertainment	Selective reporting	Other bias	Overall risk of bias	GRADE
1	**Weight, body composition**	Brekke et al. (2014) *(16)*	unclear	low	high	high	low	unclear	low	**high**	⊕
		Kinnunen et al. (2007) *(26)*	high	high	high	high	low	unclear	low	**high**	
		Tripette et al. (2014) *(15)*	unclear	unclear	high	unclear	unclear	unclear	low	**unclear**	
		O'Toole et al. (2003) *(30)*	low	low	high	unclear	low	unclear	low	**unclear**	
		Bertz et al. (2012) *(34)*	low	low	high	unclear	low	unclear	low	**unclear**	
		Osman et al. (2016) *(32)*	low	low	high	unclear	high	unclear	low	**high**	
2	**Postnatal depression**	Da Costa et al. (2009) *(27)*	unclear	low	high	unclear	unclear	unclear	low	**unclear**	⊕
		Heh et al. (2008) *(21)*	high	high	high	low	high	unclear	low	**high**	
		Lewis et al. (2014) *(24)*	low	low	high	low	low	unclear	low	**low**	
		Daley et al. (2015) *(25)*	low	low	high	unclear	low	unclear	low	**unclear**	
		Daley et al. (2008) *(28)*	low	unclear	high	unclear	low	unclear	low	**unclear**	
		Armstrong et al. (2003) *(18)*	low	low	high	unclear	low	uncear	unclear	**unclear**	
		Norman et al. (2010) *(31)*	low	low	high	unclear	low	unclear	low	**unclear**	
3	**Physical fitness**	Zourladani et al. (2012) *(37)*	high	high	high	unclear	unclear	unclear	high	**high**	⊕
		Lovelady et al. (1995) *(36)*	unclear	unclear	high	unclear	low	unclear	low	**unclear**	
4	**Fatigue**	Dritsa et al. (2008) *(17)*	unclear	low	high	unclear	unclear	unclear	low	**unclear**	⊕
		Ashrafinia et al. (2015) *(22)*	high	high	high	high	high	unclear	high	**high**	
5	**Lactation**	Dewey et al. (1994) *(38)*	unclear	unclear	high	low	low	unclear	low	**unclear**	⊕⊕⊕
		McCrory et al. (1999) *(35)*	low	unclear	high	unclear	low	unclear	low	**unclear**	
6	**Glycaemia**	Youngwanichsetha et al. (2013) *(20)*	Unclear	Low	high	Unclear	low	unclear	low	**unclear**	⊕
7	**Physical activity**	McIntyre et al. (2012) *(29)*	unclear	unclear	high	unclear	unclear	unclear	unclear	**unclear**	⊕⊕
8	**Sleep quality**	Ashrafini et al. (2015) *(22)*	high	high	high	high	high	uncear	high	**high**	⊕
10	**Infant growth**	Lovelady et al. (2000) *(33)*	low	unclear	high	unclear	low	unclear	low	**unclear**	⊕
11	**Bone density**	Lovelady et al. (2009) *(19)*	unclear	unclear	high	unclear	low	unclear	low	**unclear**	⊕⊕

GRADE score and interpretation

⊕⊕⊕⊕ **High.** We are very confident that the true effect lies close to that of the estimate of the effect.

⊕⊕⊕O **Moderate.** We are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

⊕⊕OO **Low.** Our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

⊕OOO **Very low.** We have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

Table 3. Risk of bias and GRADE assessments.

Study	Supervised/ monitored/unsupervised	LTFU intervention group(s)	Adherence to intervention	Exercise effect on primary outcome? (by CI or p<0.05)
Unsupervised studies				
Tripette et al. (2014) *(15)*	Unsupervised	Not reported	Not reported	**Yes (p<0.001)**
O'Toole et al. (2003) *(30)*	Unsupervised	4/21 (19%)	17/21 (81%)	**Yes (p<0.05)**
Lewis et al. (2014) *(24)*	Unsupervised	5/66 (8%)	Not reported	**No**
Daley et al. (2015) *(25)*	Unsupervised	4/47(8.5%)	41/47 (87%)	**Yes (p=0.03)**
Youngwanichsetha et al. (2013) *(20)*	Unsupervised	2/32 (6%)	Not reported	**Yes (p=0.02)**
Ashrafinia et al. (2015) *(22)*	Unsupervised	Not reported	Not reported	**Yes (p<0.001)**
McIntyre et al. (2012) *(29)*	Unsupervised	1/15 (7%)	6/14 (40%)	**No**
Ashrafini et al. (2015) *(22)*	Unsupervised	Not reported	Not given	**Yes (p<0.001)**
Lovelady et al. (2009) *(19)*	Unsupervised	3/10 (30%)	83.4% aerobic sessions	**No**
		14.5% (reported by 6/9)	**73.6% (reported by 4/9)**	**6/9 showed effect**
Supervised studies				
Brekke et al. (2014) *(16)*	HR monitor	1/34 (3%)	62/68 (91%)	**No**
Bertz et al. (2012) *(34)*	HR monitor	2/34 (6%)	Not reported	**No**
Da Costa et al. (2009) *(27)*	HR monitor	6/46 (13%)	35/46 (76%)	**No**
Dritsa et al. (2008) *(17)*	HR monitor	6/46 (13%)	Not reported	**Yes (p0.016)**
McCrory et al. (1999) *(35)*	HR monitor	2/44 (5%)	21/22 (95%)	**No**
Daley et al. (2008) *(28)*	Pedometer	4/20 (20%)	Not reported	**No**
Kinnunen et al. (2007) *(26)*	Unsupervised, optional supervised	5/53 (9%)	Not reported	**No**
Heh et al. (2008) *(21)*	Supervised and unsupervised	2/35 (5.7%)	Not reported	**Yes (p=0.01)**
Osman et al. (2016) *(32)*	Supervised	Not reported	Not reported	**No**
Armstrong et al. (2003) *(18)*	Supervised, Pedometer	Not reported	64% attendance	**Yes (p<0.01)**
Norman et al. (2010) *(31)*	Supervised	2/62 (3%)	130/130 (100%)	**No**
Zourladani et al. (2012) *(37)*	Supervised	Not reported	Not reported	**Yes (p<0.001)**
Lovelady et al. (1995) *(36)*	Supervised	Not reported	Not reported	**Yes (p<0.0001)**
Dewey et al. (1994) *(38)*	Supervised	Not reported	149/165 sessions completed (90%)	**No**
Lovelady et al. (2000) *(33)*	Supervised	6/27 (22%)	Not given	**No**
		9.97% (reported by 10/15)	**86% (reported by 6/15)**	**5/15 showed effect**

Table 4. Comparison of studies using unsupervised and supervised interventions for mean % lost to follow-up (LTFU), mean % adherence and effect of interventions on primary outcome.

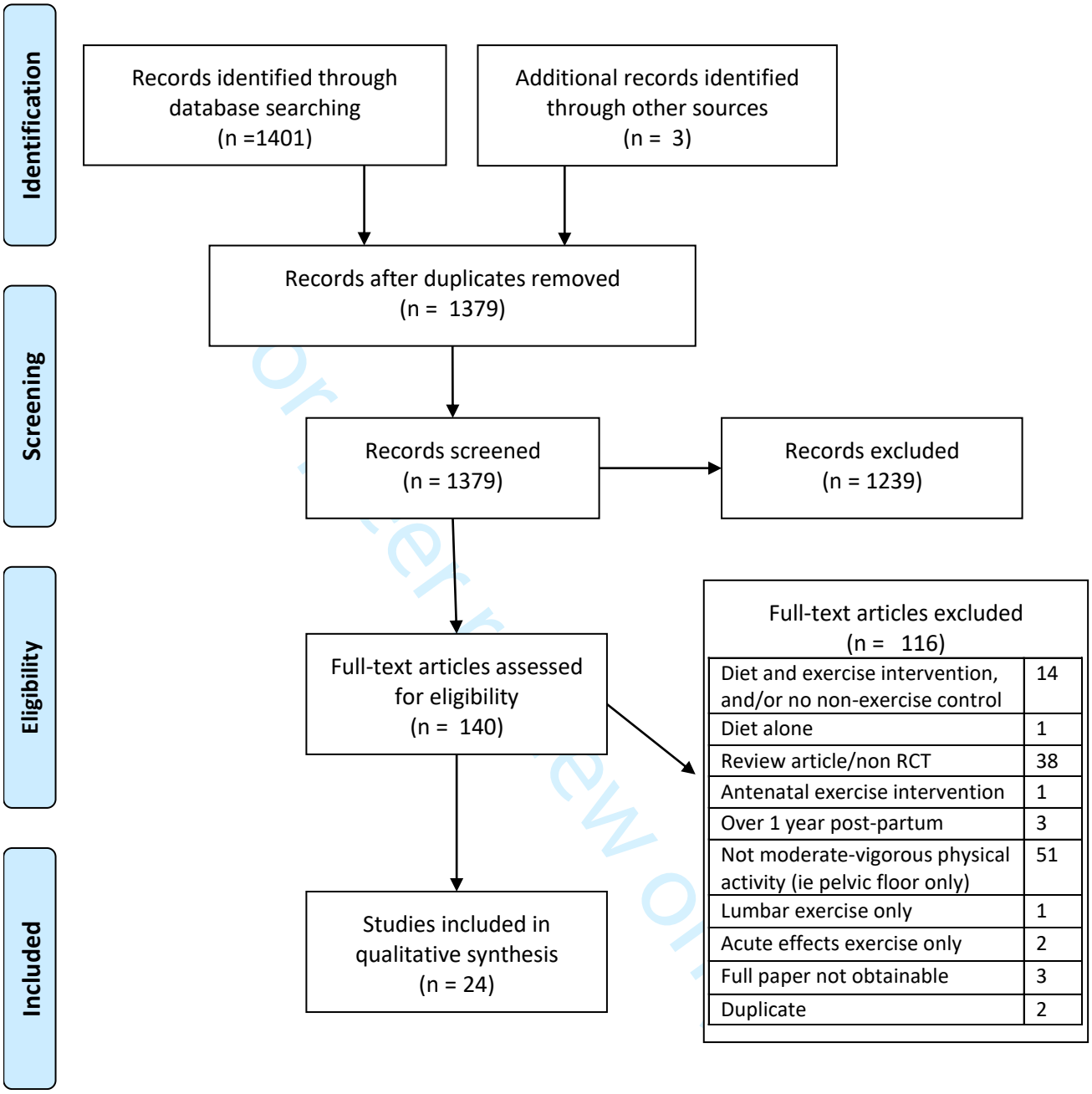


Fig. 1 PRISMA diagram for Identification, Screening, Eligibility and Included reports.

Medline search

1. postpartum period/ or lactation/
2. postpartum period.mp.
3. Peripartum Period/
4. peripartum period.mp.
5. postpartum women.mp.
6. post partum women.mp.
7. (post natal or postnatal).tw.
8. (postpartum or post partum).tw.
9. (post pregnancy or postpregnancy).tw.
10. ((Post or after or follow*) adj birth*).tw.
11. ((Post or after or follow*) adj labo?r).tw.
12. ((Post or after or follow*) adj childbirth*).tw.
13. ((Post or after or follow*) adj pregnancy).tw.
14. ((Post or after or follow*) adj parturition).tw.
15. ((Post or after or follow*) adj delivery).tw.
16. (interpregnanc* or inter pregnanc* or (between adj pregnanc*)).tw.
17. (interconception or inter conception or (between adj conception)).tw.
18. puerperium.tw.
19. or/1-18
20. exercise/ or cool-down exercise/ or gymnastics/ or muscle stretching exercises/ or physical conditioning, human/ or circuit-based exercise/ or high-intensity interval training/ or plyometric exercise/ or resistance training/ or running/ or jogging/ or swimming/ or walking/ or warm-up exercise/ or pronation/ or supination/
21. exercise.tw

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22. exercise therapy/ or motion therapy, continuous passive/

23. motor activity/ or physical exertion/ or physical fitness/ or cardiorespiratory fitness/ or postural balance/

24. sports/ or athletic performance/ or baseball/ or basketball/ or bicycling/ or boxing/ or football/ or golf/ or gymnastics/ or hockey/ or martial arts/ or tai ji/ or mountaineering/ or racquet sports/ or tennis/ or return to sport/ or running/ or jogging/ or skating/ or snow sports/ or skiing/ or soccer/ or sports for persons with disabilities/ or "track and field"/ or volleyball/ or walking/ or water sports/ or swimming/ or diving/ or weight lifting/ or wrestling/ or youth sports/

25. sports medicine/ or sports nutritional sciences/

26. fitness.tw.

27. (activit* and program*).tw.

28. (locomotor and activit*).tw.

29. (motor and activit*).tw.

30. (structure* and activit*).tw.

31. training*.tw.

32. (therap* and exercis*).tw.

33. physical*.tw.

34. (activit* or conditioning* or effort* or exert*).tw.

35. 33 and 34

36. 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 35

37. randomized controlled trial'.pt.

38. (random$ or placebo$ or single blind$ or double blind$ or triple blind$).mp.

39. (retraction of publication or retracted publication or controlled clinical trial).pt. or randomi?ed.tw. or randomly.tw.

40. controlled clinical trial.pt.

41. randomi?ed.tw.

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42. randomly.tw.

43. or/37-42

44. exp animals/ not humans.sh.

45. (random sampl$ or random digit$ or random effect$ or random survey or random regression).mp. not 'randomized controlled trial'.pt.

46. 43 not (44 or 45)

47. 19 and 36 and 46

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Embase search strategy

1. postnatal care/ or lactation/
2. postnatal care.mp.
3. perinatal period/
4. perinatal period.mp.
5. postpartum period.mp.
6. peripartum period.mp.
7. postpartum women.mp.
8. post partum women.mp.
9. (postnatal or post natal).ti,ab.
10. (post pregnancy or postpregnancy).ti,ab.
11. ((Post or after or follow*) adj birth*).ti,ab.
12. ((Post or after or follow*) adj labo?r).ti,ab.
13. ((Post or after or follow*) adj childbirth*).ti,ab.
14. ((Post or after or follow*) adj pregnancy).ti,ab.
15. ((Post or after or follow*) adj parturition).ti,ab.
16. ((Post or after or follow*) adj delivery).ti,ab.
17. (interpregnanc* or inter pregnanc* or (between adj pregnanc*)).ti,ab.
18. (interconception or inter conception or (between adj conception)).ti,ab.
19. puerperium/
20. puerperium.ti,ab.
21. or/1-20
22. exercise/ or aerobic exercise/ or anaerobic exercise/ or aquatic exercise/ or arm exercise/ or athletic performance/ or breathing exercise/ or circuit training/ or closed kinetic chain exercise/ or cool down/ or dynamic exercise/ or endurance

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training/ or high intensity interval training/ or isokinetic exercise/ or leg exercise/ or muscle exercise/ or open kinetic chain exercise/ or pilates/ or plyometrics/ or resistance training/ or static exercise/ or warm up/

23. exercise.ti,ab.

24. kinesiotherapy/ or pelvic floor muscle training/ or stretching exercise/

25. athletic performance/

26. sport/ or aquatic sport/ or athletics/ or baseball/ or basketball/ or body building/ or combat sport/ or contact sport/ or "cricket (sport)"/ or cycling/ or diving/ or endurance sport/ or extreme sport/ or football/ or golf/ or hockey/ or horseback riding/ or ice hockey/ or jogging/ or motor sport/ or racquet sport/ or rock climbing/ or roller skating/ or rowing/ or rugby/ or running/ or skateboarding/ or skating/ or skiing/ or soccer/ or swimming/ or tai chi/ or team sport/ or triathlon/ or volleyball/ or water skiing/ or winter sport/ or yoga/ or youth sport/

27. sports medicine/

28. fitness/

29. fitness.ti,ab.

30. (activit* and program*).ti,ab.

31. (locomotor and activit*).ti,ab.

32. (motor and activit*).ti,ab.

33. (structure* and activit*).ti,ab.

34. training*.ti,ab.

35. (therap* and exercis*).ti,ab.

36. physical*.ti,ab.

37. (activit* or conditioning* or effort* or exert*).ti,ab.

38. 36 and 37

39. 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 38

40. (random$ or placebo$ or single blind$ or double blind$ or triple blind$).ti,ab

41. RETRACTED ARTICLE/

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42. randomi?ed.ti,ab.

43. randomly.ti,ab.

44. or/40-43

45. exp animal$/ not human$.sh,hw.

46. (random sampl$ or random digit$ or random effect$ or random survey or random regression).ti,ab. not exp randomized controlled trial/

47. 44 not (45 or 46)

48. 21 and 39 and 47

49. limit 48 to embase

50. limit 49 to exclude medline journals

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CENTRAL SEARCH STRATEGY

1. MeSH descriptor: [Postpartum Period] explode all trees
2. MeSH descriptor: [Lactation] explode all trees
3. (postpartum period)
4. MeSH descriptor: [Peripartum Period] explode all trees
5. (peripartum period)
6. (postpartum women)
7. (post partum women)
8. ((post natal or postnatal)):ti,ab,kw
9. ((post partum or postpartum)):ti,ab,kw
10. ((post pregnancy or postpregnancy)):ti,ab,kw
11. (((Post or after or follow*) NEXT birth*)):ti,ab,kw
12. (((Post or after or follow*) NEXT labo?r)):ti,ab,kw
13. (((Post or after or follow*) NEXT childbirth*)):ti,ab,kw
14. (((Post or after or follow*) NEXT pregnancy)):ti,ab,kw
15. (((Post or after or follow*) NEXT parturition)):ti,ab,kw
16. (((Post or after or follow*) NEXT delivery)):ti,ab,kw
17. ((interpregnanc* or inter pregnanc* or (between NEXT pregnanc*))):ti,ab,kw
18. ((interconception or inter conception or (between NEXT conception))):ti,ab,kw
19. (puerperium):ti,ab,kw
20. #1 or #2 or #3 or #4 or #5 #8 or #9 or #10 or #11 or #12 or #13 or #14 or #15 or #16 or #17 or #18 or #19
21. MeSH descriptor: [Exercise] explode all trees
22. (exercise):ti,ab,kw

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23. MeSH descriptor: [Exercise Therapy] explode all trees
24. MeSH descriptor: [Physical Fitness] explode all trees
25. MeSH descriptor: [Sports] explode all trees
26. MeSH descriptor: [Sports Medicine] this term only
27. MeSH descriptor: [Sports Nutritional Sciences] this term only
28. (fitness):ti,ab,kw
29. ((activit* and program*)):ti,ab,kw
30. ((locomotor and activit*)):ti,ab,kw
31. ((motor and activit*)):ti,ab,kw
32. ((structure* and activit*)):ti,ab,kw
33. (training*):ti,ab,kw
34. ((therap* and exercis*)):ti,ab,kw
35. (physical*):ti,ab,kw
36. ((activit* or conditioning* or effort* or exert*)):ti,ab,kw
37. #35 and #36
38. #21 or #22 or #23 or #24 or #25 or #26 or #27 or #28 or #29 or #30 or #31 or #32 or #33 or #34 or #37
39. #20 and #38

1401

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MOOSE (Meta-analyses Of Observational Studies in Epidemiology) Checklist

A reporting checklist for Authors, Editors, and Reviewers of Meta-analyses of Observational Studies. You must report the page number in your manuscript where you consider each of the items listed in this checklist. If you have not included this information, either revise your manuscript accordingly before submitting or note N/A.

Reporting Criteria	Reported (Yes/No)	Reported on Page No.
Reporting of Background		
Problem definition		
Hypothesis statement		
Description of Study Outcome(s)		
Type of exposure or intervention used		
Type of study design used		
Study population		
Reporting of Search Strategy		
Qualifications of searchers (eg, librarians and investigators)		
Search strategy, including time period included in the synthesis and keywords		
Effort to include all available studies, including contact with authors		
Databases and registries searched		
Search software used, name and version, including special features used (eg, explosion)		
Use of hand searching (eg, reference lists of obtained articles)		
List of citations located and those excluded, including justification		
Method for addressing articles published in languages other than English		
Method of handling abstracts and unpublished studies		
Description of any contact with authors		
Reporting of Methods		
Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested		
Rationale for the selection and coding of data (eg, sound clinical principles or convenience)		
Documentation of how data were classified and coded (eg, multiple raters, blinding, and interrater reliability)		
Assessment of confounding (eg, comparability of cases and controls in studies where appropriate		

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Reporting Criteria	Reported (Yes/No)	Reported on Page No.
Assessment of study quality, including blinding of quality assessors; stratification or regression on possible predictors of study results		
Assessment of heterogeneity		
Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated		
Provision of appropriate tables and graphics		
Reporting of Results		
Table giving descriptive information for each study included		
Results of sensitivity testing (eg, subgroup analysis)		
Indication of statistical uncertainty of findings		
Reporting of Discussion		
Quantitative assessment of bias (eg, publication bias)		
Justification for exclusion (eg, exclusion of non–English-language citations)		
Assessment of quality of included studies		
Reporting of Conclusions		
Consideration of alternative explanations for observed results		
Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)		
Guidelines for future research		
Disclosure of funding source		

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POSTNATAL EXERCISE INTERVENTIONS: A SYSTEMATIC REVIEW OF ADHERENCE AND EFFECT

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POSTNATAL EXERCISE INTERVENTIONS: A SYSTEMATIC REVIEW OF ADHERENCE AND EFFECT

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ABSTRACT

Objective to evaluate adherence to and effect of postnatal physical activity (PA) interventions.

Design systematic review of PA intervention randomised controlled trials in postnatal women. The initial search was carried out in September 2018, and updated in January 2021.

Data sources Embase, MEDLINE, and Cochrane Central Register of Controlled Trials (CENTRAL) databases, hand-searching references of included studies. The 25 identified studies included 1466 postnatal women in community and secondary care settings.

Eligibility criteria studies were included if the PA interventions were commenced and assessed in the postnatal year.

Data extraction and synthesis data was extracted using a pre-specified extraction template and assessed independently by two reviewers using Cochrane ROB 1 tool.

Results 1413 records were screened for potential study inclusion, full-text review was performed on 146 articles, 25 studies were included. The primary outcome was adherence to PA intervention. The secondary outcomes were the effect of the PA interventions on the studies' specified primary outcome. We compared effect on primary outcome for supervised and unsupervised exercise interventions. Studies were small, median *N*= 66 (20-130). PA interventions were highly variable, targets for PA per week ranged from 60 -275 minutes per week. LTFU was higher (14.5% vs 10%) and adherence to intervention was lower (73.6% vs. 86%) for unsupervised vs. supervised studies.

Conclusions studies of PA interventions inconsistently reported adherence and LTFU. Where multiple studies evaluated PA as an outcome, they had inconsistent effects, with generally low study quality and high risk of bias. Agreement for effect between studies was evident for PA improving physical fitness and reducing fatigue. Three studies showed no adverse effect of physical activity on breast feeding. High-quality research reporting adherence and LTFU is needed into how and when to deliver postnatal PA interventions to benefit postnatal physical and mental health.
Registration number PROSPERO CRD42019114836

Strengths and Limitations

- This review considered only PA intervention studies which were compared to a control group with no PA intervention or with advice to be sedentary, to allow evaluation of effect of the PA intervention on the primary outcomes of each study.
- There may be studies which compare PA interventions in groups other than postnatal women which had high adherence and positive or negative effects and will have been excluded from our analysis.
- We considered only PA interventions in the year after delivery, potentially excluding interventions later after delivery which may be appealing to women and effective in improving physical and mental health.

INTRODUCTION

Physical inactivity is estimated to cause between 6 and 10 percent (%) of the major non-communicable diseases of breast and colon cancer, type 2 diabetes, and coronary heart disease; and 9% of premature mortality worldwide (1). Not surprisingly, physical inactivity is responsible for 1 in 6 UK deaths and is

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estimated to cost the UK £7.4 billion annually (2). In the UK 42% of women are estimated to be not active enough to maintain good health (2).

Adult physical activity (PA) guidelines recommend 150 minutes of moderate-to-vigorous intensity physical activity (MVPA) per week; only 67% of UK women aged 35-44 achieve this with further reductions with age (3). Disparity also exists from a young age; the percentage of girls aged 11 achieving the recommended 60 minutes of daily MVPA declines from 20% to 12% by the age of 15; in boys this falls from 32% to 25% (4).

Women's physical activity levels are low during and after pregnancy. 60% of women during pregnancy report engaging in no leisure time PA (5). Fewer than 13% of women meet recommended PA levels during pregnancy, and this decreases with each trimester (6). PA measured during and after pregnancy is predominately of a low intensity, providing somewhat limited health benefits. It is important to note that postnatal total PA (measured in hours or weeks) is similar to that seen in mid pregnancy, but comprises a larger proportion of care-giving activities (7).

The postnatal time is a difficult period for women to restart or maintain PA. Fatigue from poor sleep and the demands of infant feeding can also contribute to lower PA levels. The Institute of Health Visitors advise waiting for the 6-week postnatal visit with women's general practitioners (GPs) before resuming high-impact exercise. GPs have to assess maternal and child health at this visit, though this visit is generally short and advice given can be extremely variable (8,9). Healthcare professionals need to know which interventions they can best recommend to post-partum women, enabling them to meet recommended levels of PA and optimise their health.

The UK Chief Medical Officers recommend that pregnant and postnatal women undertake 150 minutes moderate PA per week, and have provided useful infographics which are available as a guide for women and healthcare professionals (10,11). These are endorsed by the American College of Obstetricians and Gynaecologists' (ACOG) guidelines, who also recommend 20 to 30 minutes of PA per day during pregnancy on most or all days of the week (12), and briefly describe the postpartum period as an opportune time to gradually resume an exercise routine. Clearly however, the figures above indicate that this is not being achieved.

To review available studies on postnatal physical activity interventions, we conducted a systematic review to determine adherence to postnatal physical activity interventions, and the effect of the interventions on the outcomes investigated in the studies identified. The results would also inform healthcare professionals who advise women on the most efficacious PA interventions for desired effect.

We wished to explore whether adherence to PA interventions would be greater for supervised (in person supervision or with monitoring of PA e.g. a pedometer) vs. unsupervised interventions; and that the effects of PA interventions on the primary outcomes of studies would be related to adherence.

METHODS

PPI PPI for a separate study helped to conceive the research question for this study. We did not undertake PPI for this study.

We carried out a systematic review based on a pre-published protocol (PROSPERO CRD42019114836) and reported the findings in accordance with specifications recommended by the per the PRISMA checklist (13).

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Search strategy and selection criteria Embase, MEDLINE, and Cochrane Central Register of Controlled Trials (CENTRAL) databases were searched in duplicate as per the search strategies (supplementary files 1, 2 and 3), without restrictions on publication date. The references of included articles were also hand-searched by the reviewers to highlight further studies for potential inclusion. This review was performed in September 2018, and an updated search was carried out in January 2021. The updated literature research identified one additional report (14) which included the same participants as those reported in (15).

An initial screen of titles and abstracts was performed independently by two reviewers (SS and EM), to identify articles for further review. Articles were included after full text review if they were randomised, controlled trials which compared moderate and/or vigorous PA to no PA. This could be in combination with another intervention e.g. a dietary intervention, if a comparison group with the same dietary intervention alone was tested. If more than one intervention was included, it was essential that there was a comparison between exercise and no exercise groups. If there was any ambiguity regarding a study's inclusion, a third reviewer (AM) decided on eligibility. Where data was missing or incomplete, contact with the corresponding author was made by email.

Where the PA intervention was restricted to pelvic floor exercises, physiotherapy or single exercise sessions, studies were excluded. Studies were excluded from the review if they did not report on original data or were animal studies. Studies were not excluded based on language, study type or publication status. There was no other restriction on the type of PA intervention.

Data analysis Two reviewers (SS and EM) extracted the data from the included studies using a pre-specified extraction template. Extracted data elements recorded for each study included general information, eligibility criteria, study characteristics, (setting, country, year, design, type and duration of PA intervention, sample size), adherence to the intervention, and primary outcome measures.

Both reviewers assessed risk of bias for each study independently using the Cochrane Collaboration's tool for assessing risk of bias (16). The following domains were assessed as having a high, low or unclear risk of bias: random sequence generation, allocation concealment, blinding of participants/ personnel, blinding of outcome, incomplete outcome, selective reporting and overall bias. The reviewers then used the GRADE criteria (13) to assess the quality of the evidence within the review for the specific outcomes identified.

RESULTS

After the initial search (Supplemental files 1,2 and 3), 1413 records were screened for potential study inclusion. After title and abstract review, full-text review was performed on 146 articles. Of these, 25 articles reporting the effect of randomised controlled trials of postnatal exercise interventions vs. no exercise were included (Table 1). Study designs included; control vs. exercise; control vs. dietary intervention vs. dietary intervention and exercise; and control vs. dietary intervention vs. exercise intervention vs. dietary and exercise intervention (Table 1). The 25 studies included 1466 postnatal women.

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	Outcome category	Author	Setting	Country	*N*	Design	Intervention group 1	Intervention group 2	Duration intervention	Supervised/ monitored/ unsupervised	PA target per week	Comparison group 1	Comparison group 2	Primary outcome measure
1	**Weight, Body composition**	Brekke et al. (2014) *(16)*	Community	Sweden	*68*	C, D, E or DE.	45 min brisk walk 60-70% MHR*4/week monitored by HR monitor and diary.	as for group 1 +dietary intervention	6 months	HR monitor	180 mins	Usual care	Dietary intervention	Waist circumference (cm)
		Kinnunen et al. (2007) *(26)*	Community	Finland	*85*	C, E	Counselling sessions *5 at 2, 3, 5 6 and 10 months postnatal. Option of attending weekly supervised exercise class.		8 months	Unsupervised, optional supervised	150 mins	Usual care		% return to normal weight at 10 months postnatal
		Tripette et al. (2014) *(15)*	Community	Japan	*34*	C, E	Active video games for 30 mins day (WiiFit and balance board)		40 days	Unsupervised	Not reported	Usual care		Weight loss kg
		O'Toole et al. (2003) *(30)*	Community	USA	*40*	D, DE	Group educational sessions *17-23 on PA and diet, combined effect 500kcal daily energy deficit		6-10 months	Unsupervised	Energy deficit of 1050kcal	Dietary counselling session *1, 500kcal/ day energy defecit		Weight loss kg
		Bertz et al. (2012) *(34)*	Community	Sweden	*68*	C, D, E or DE.	E 45 min brisk walk 60-70% MHR*4/week monitored by HR monitor and diary	as for group 1 +dietary intervention	12 weeks	HR monitor	180 mins	Usual care	Dietary intervention 500 kcal/day deficit	Weight loss kg
		Osman et al. (2016) *(32)*	Community	Egypt	*47*	D, DE	Supervised aerobic exercise sessions *3/week for 4 weeks to 60-70% MHR, dietary and BF advice		4 weeks	Supervised	105 mins	Dietary and BF advice		Weight loss kg
2	**Postnatal depression**	Da Costa et al. (2009) *(27)*	Community	Canada	*88*	C, E	Counselling sessions *4 with exercise physiologist over 12 weeks. 60-120mins/week at 60-85% MHR		12 weeks	HR monitor	120 mins	Usual care		EPDS

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		Heh et al. (2008) *(21)*	Community	Taiwan	*80*	C, E	Exercise program CD for 2 home exercise sessions/ week, 1 hour weekly exercise session at hospital	3 months	Supervised and unsupervised	180 mins	Usual care	EPDS
		Lewis et al. (2014) *(24)*	Community	USA	*130*	C, E	11 telephone sessions to help women increase MVPA,	6 months	Unsupervised, Actigraph 7 days	Not reported	Usual care	% depression by SCID-1
		Daley et al. (2015) *(25)*	Community	England	*94*	C, E	2* exercise counselling sessions, 2* phone calls, leaflets months 3,4,5 and 6	6 months	Unsupervised	150 mins	Usual care	EPDS
		Daley et al. (2008) *(28)*	Primary care, community	England	*38*	C, E	2* exercise counselling sessions, 2* phone calls	12 weeks	Pedometer	105 mins	Usual care	EPDS
		Armstrong et al. (2003) *(18)*	Community	Australia	*20*	C, E	Group Pram walking 3*/week, 30–40 mins. at 60–75% of MHR, weekly social support session	12 weeks	Supervised, Pedometer	90-120 mins	Usual care	EPDS
		Norman et al. (2010) *(31)*	Community	Australia	*161*	C, E	1*exercise class/week. 1* General health advice education session	4 weeks	Supervised	60 mins	1* General health education session	Positive affect balance scale
3	**Physical fitness**	Zourladani et al. (2012) *(37)*	Home	Greece	*37*	C, E	3* exercise classes/week, 50-60mins aerobic and strengthening exercise	12 weeks	Supervised	150-180 mins	Usual care	VO2max (ml/kg/min)
		Lovelady et al. (1995) *(36)*	Community	USA	*33*	C, E	5* exercise sessions/week, 45 mins aerobic exercise sessions	12 weeks	Supervised	225mins	Usual care	VO2max (ml/kg/min)
4	**Fatigue**	Dritsa et al. (2008) *(17)*	Community	Canada	*88*	C, E	4*Counselling sessisons to produce exercise prescription.	12 weeks	HR monitor	120mins	Usual care	Multidimensional Fatigue Inventory (MFI-20) -general fatigue

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		Ashrafinia et al. (2015) *(22)*	Home	Iran	*80*	C, E	4 antenatal pilates training sessions, 5*home pilates sessions/week, 30 mins, exercise leaflet and CD		8 weeks	Unsupervised	150 mins			(MFI-20) - general fatigue
5	**Lactation**	Dewey et al. (1994) *(38)*	Community	USA	*33*	C, E	5* exercise sessions/week, individually tailored, gradated increase to 45 mins,		12 weeks	Supervised	225mins	Usual care		Infant milk intake g/kg/day
		McCrory et al. (1999) *(35)*	Community	USA	*67*	C, D, or DE.	Exercise 9/11 days at 50-70% maximum HR	as for group 1 +dietary intervention	11 days	HR monitor	Not reported	Usual care	Dietary intervention	Milk volume
6	**Glycaemia**	Youngwanichsetha et al. (2013) *(20)*	Hospital	Thailand	*64*	C, E	3*tai chi qigong sessions/week, 50mins		12 weeks	Unsupervised	150 mins	Usual care		Fasting plasma glucose (mg/dl)
7	**Physical activity**	McIntyre et al. (2012) *(29)*	Community	Australia	*28*	C, E	1*face to face exercise consultation, 8 telephone calls		12 weeks	Unsupervised	150 mins	Usual care		Change in PAmins/week (median)
8	**Sleep quality**	Ashrafini et al. (2015) *(22)*	Home	Iran	*80*	C, E	4 antenatal pilates training sessions, 5*home pilates sessions/week, 30 mins, exercise leaflet and CD		8 weeks	Unsupervised	150 mins	1*postnatal care advice session		Pittsburg Sleep Index (global score)
10	**Infant growth**	Lovelady et al. (2000) *(33)*	Community	USA	*48*	D, DE	4*exercise sessions/week, 45 mins, at 65-80% MHR		10 weeks	Supervised	180 mins	Usual care		Gain in Infant Weight (g)
11	**Bone density**	Lovelady et al. (2009) *(19)*	Home	USA	*20*	C, E	6*exercise sessions/week, 45 mins, 65-80% MHR, aerobic and resistance		16 weeks	Unsupervised	270 mins	Usual care		Change in whole body bone mineral density (g/cm2)

Key: C-control, E-Exercise intervention, D-dietary intervention, DE-ietary and exercise intervention, HR-heart rate, MHR-maxiumum heart rate, CV-cardiovascular, MVPA moderate or vigorous physical activity, PA -physical activity, FU-follow-up, BF-breast feeding

Table 1. Study characteristics, interventions, comparison groups and outcomes

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Studies included were conducted in North America, Europe, Africa, Asia and Australasia. Studies were generally small, ranging from 20 to 130 participants, median *N*=66. PA interventions used were varied, including provision of active video games (WiiFit and balance board, (17)), exercise prescriptions (18–22), CDs and DVDs (23–25), telephone (26,27) and face-to-face counselling sessions (27–31), group classes (24,25,32,33) and supervised exercise sessions (33,15,34,14), . Interventions varied in duration between 11 days and 10 months. Where reported, the targets for PA per week ranged from 60 to 275 minutes per week. One study prescribed exercise to attain an individualised energy deficit. Four studies did not report a PA target. A meta-analysis was not performed due to the heterogeneity of the studies identified.

Of the studies included, 9 used unsupervised exercise interventions (17,21,22,24–27,31,32), and 16 used exercise with some form of supervision. 6 studies used PA monitoring (a pedometer or an Actigraph, (18,19,29,30,35,36)), 1 used PA monitoring and in-person supervised exercise interventions (20), 6 used in-person supervised interventions (33,15,34,37–39,14), and 2 used a mixture of in-person supervised and unsupervised interventions (23,28). (One study used an actigraph for 7 days prior to follow-up 6 month post intervention though not during the intervention (23).)
Loss to follow-up (LTFU)in the exercise intervention group was reported in 17/26 studies (18,19,21–23,26–36)and adherence to the PA intervention was reported by 11/26 studies (18,20,21,27,29,31–33,36,39). When reported, for the groups undergoing PA interventions, mean LTFU was 11.8% and mean adherence was 80.4%. Some studies reported only overall rates of LTFU and adherence. In studies where LTFU and adherence is reported, the proportions were similar for intervention and control/comparison groups.

Effect of PA intervention on weight loss was reported by 2/6 studies (17,32), with weight loss between 2.1 and 5.6 kg reported (Table 2). In studies which compared dietary, PA and combined interventions, PA interventions were not associated with weight loss over and above the effect of the dietary intervention (18,35). Higher rates of LTFU and lower rates of adherence (where reported) were seen in studies which showed an effect on weight loss.

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	Outcome category	Author	Primary outcome measure	LTFU intervention group(s)	Adherence to intervention	Effect in Intervention group 1	Effect in Intervention group 2	Comparison group 1	Comparison group 2	Exercise effect on primary outcome? (by CI or p<0.05)
1	**Weight, Body composition**	Brekke et al. (2014) *(18)*	Waist circumference (cm)	1/34 (3%)	62/68 (91%)	-4.3	-8.8	-1.3	-8.7	**No**
		Kinnunen et al. (2007) *(28)*	% return to normal weight at 10 months postnatal	5/53 (9%)	Not reported	50		30		**No**
		Tripette et al. (2014) *(17)*	Weight loss kg	Not reported	Not reported	-2.2		-0.5		**Yes (p<0.001)**
		O'Toole et al. (2003) *(32)*	Weight loss kg	4/21 (19%)	17/21 (81%)	-5.6		-0.7		**Yes (p<0.05)**
		Bertz et al. (2012) *(35)*	Weight loss kg	2/34 (6%)	Not reported	-2.4	-6.9	-0.8	-8.3	**No**
		Osman et al. (2016) *(15)*	Weight loss kg	Not reported	Not reported	-1.56		+0.09		**No**
2	**Postnatal depression**	Da Costa et al. (2009) *(27)*	Ediburgh Postnatal Depression Scale (EPDS)	6/46 (13%)	35/46 (76%)	-4.6		-5		**No**
		Heh et al. (2008) *(21)*	EPDS	2/35 (5.7%)	Not reported	-6.4		-3.5		**Yes (p=0.01)**
		Lewis et al. (2014) *(24)*	% depression by SCID-1	5/66 (8%)	Not reported	8.2		7.94		**No**
		Daley et al. (2015) *(25)*	EPDS	4/47(8.5%)	41/47 (87%)	-4.79		-2.83		**Yes (p=0.03)**
		Daley et al. (2008) *(28)*	EPDS	4/20 (20%)	Not reported	-4.9		-4.6		**No**
		Armstrong et al. (2003) *(18)*	EPDS	Not reported	64% attendance	-12.8		-3.7		**Yes (p<0.01)**
		Norman et al. (2010) *(31)*	Positive affect balance scale	2/62 (3%)	130/130 (100%)	+1.19		-0.18		**No**

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3	**Physical fitness**	Zourladani et al. (2012) *(37)*	VO2max (ml/kg/min)	Not reported	Not reported	not reported	not reported		**Yes (p<0.001)**
		Lovelady et al. (1995) *(36)*	VO2max (ml/kg/min)	Not reported	Not reported	+6.8	+1.3		**Yes (p<0.0001)**
4	**Fatigue**	Dritsa et al. (2008) *(17)*	MFI-20 -general fatigue	6/46 (13%)	Not reported	-3.87	-1.83		**Yes (p0.016)**
		Ashrafinia et al. (2015) *(22)*	MFI-20 -general fatigue	Not reported	Not reported	-3.35	+1.2		**Yes (p<0.001)**
5	**Lactation**	Dewey et al. (1994) *(38)*	Infant milk intake g/kg/day	Not reported	149/165 sessions completed (90%)	117	118		**No**
		McCrory et al. (1999) *(35)*	Milk volume g/day	2/44 (5%)	21/22 (95%)	-16	+17	-1	**No**
		Osman et al. (2020) *(14)*	Breast milk cortisol mcg/ml	Not reported	Not reported	-93.84	-91.68		**No**
6	**Glycaemia**	Youngwanichsetha et al. (2013) *(20)*	Fasting plasma glucose (mg/dl)	2/32 (6%)	Not reported	-10.25	-2.09		**Yes (p=0.02)**
7	**Physical activity**	McIntyre et al. (2012) *(29)*	Change in PA mins/week (median)	1/15 (7%)	6/14 (40%)	+60	+0		**No**
8	**Sleep quality**	Ashrafini et al. (2015) *(22)*	Pittsburg Sleep Index (global score)	Not reported	Not reported	5.45	8.35		**Yes (p<0.001)**
10	**Infant growth**	Lovelady et al. (2000) *(33)*	Gain in Infant Weight (g)	6/27 (22%)	Not reported	1925	1861		**No**
11	**Bone density**	Lovelady et al. (2009) *(19)*	Change in whole body bone mineral density (g/cm2)	3/10 (30%)	83.4% aerobic sessions	-0.8	-0.6		**No**

Key: EPDS - Edinburgh Postnatal Depression Scale, PA- Physical activity, MFI-20- Multidimensional Fatigue Inventory

Table 2. Study outcomes, Lost to follow-up, Adherence to interventions, effect of physical activity intervention studies on outcomes and adherence

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For measures of postnatal depression, 3/7 reported an effect of a PA intervention (20,23,27). The effect size varied greatly, from -4.79 to -12.8 points on the Edinburgh Postnatal depression scale. 2 out of 3 of these studies reported LTFU, 8% in both; adherence was reported by 2/3 and was 64 and 87%.

Physical fitness, defined by VO_2max and investigated by 2 small studies, is improved by PA interventions (37,38). General fatigue, part of the Multidimensional Fatigue Inventory, appears to be reduced by PA interventions (19,25). Lactation is not impaired by PA interventions in studies with 90 and 95% adherence (36,39), nor is infant growth (34). Glycaemia (22) and sleep quality (24) appear to be improved by PA interventions. Bone density (21), breast milk cortisol (14)and (surprisingly) physical activity (31) were not significantly affected by PA interventions, although the latter study had n=29 and an adherence of 40% to the intervention.

Risk of bias assessment found that most studies were at high (7/25) or unclear (16/25) risk of bias (Supplementary File 1, Table 3). Blinding measures for personnel and participants were absent for all studies, conceding that it would be difficult to do so, especially for supervised exercise interventions. No studies referred to a published protocol and/or was convincing that the published report included all expected outcomes, including those that were pre-specified. Many studies did not list a primary outcome, in these cases we used the title or outcome reported first or with most detail in the results section.

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	Outcome category	Study	Random sequence generation	Allocation concealment	Blinding participants/ personnel	Blinding outcome	Incomplete outcome ascertainment	Selective reporting	Other bias	Overall risk of bias	GRADE
1	**Weight, body composition**	Brekke et al. (2014) *(16)*	unclear	low	high	high	low	unclear	low	**high**	⊕
		Kinnunen et al. (2007) *(26)*	high	high	high	high	low	unclear	low	**high**	
		Tripette et al. (2014) *(15)*	unclear	unclear	high	unclear	unclear	unclear	low	**unclear**	
		O'Toole et al. (2003) *(30)*	low	low	high	unclear	low	unclear	low	**unclear**	
		Bertz et al. (2012) *(34)*	low	low	high	unclear	low	unclear	low	**unclear**	
		Osman et al. (2016) *(32)*	low	low	high	unclear	high	unclear	low	**high**	
2	**Postnatal depression**	Da Costa et al. (2009) *(27)*	unclear	low	high	unclear	unclear	unclear	low	**unclear**	⊕
		Heh et al. (2008) *(21)*	high	high	high	low	high	unclear	low	**high**	
		Lewis et al. (2014) *(24)*	low	low	high	low	low	unclear	low	**low**	
		Daley et al. (2015) *(25)*	low	low	high	unclear	low	unclear	low	**unclear**	
		Daley et al. (2008) *(28)*	low	unclear	high	unclear	low	unclear	low	**unclear**	
		Armstrong et al. (2003) *(18)*	low	low	high	unclear	low	uncear	unclear	**unclear**	
		Norman et al. (2010) *(31)*	low	low	high	unclear	low	unclear	low	**unclear**	
3	**Physical fitness**	Zourladani et al. (2012) *(37)*	high	high	high	unclear	unclear	unclear	high	**high**	⊕
		Lovelady et al. (1995) *(36)*	unclear	unclear	high	unclear	low	unclear	low	**unclear**	

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4	**Fatigue**	Dritsa et al. (2008) *(17)*	unclear	low	high	unclear	unclear	unclear	low	**unclear**	⊕
		Ashrafinia et al. (2015) *(22)*	high	high	high	high	high	unclear	high	**high**	
5	**Lactation**	Dewey et al. (1994) *(38)*	unclear	unclear	high	low	low	unclear	low	**unclear**	⊕⊕⊕
		McCrory et al. (1999) *(35)*	low	unclear	high	unclear	low	unclear	low	**unclear**	
6	**Glycaemia**	Youngwanichsetha et al. (2013) *(20)*	Unclear	Low	high	Unclear	low	unclear	low	**unclear**	⊕
7	**Physical activity**	McIntyre et al. (2012) *(29)*	unclear	unclear	high	unclear	unclear	unclear	unclear	**unclear**	⊕⊕
8	**Sleep quality**	Ashrafini et al. (2015) *(22)*	high	high	high	high	high	uncear	high	**high**	⊕
10	**Infant growth**	Lovelady et al. (2000) *(33)*	low	unclear	high	unclear	low	unclear	low	**unclear**	⊕
11	**Bone density**	Lovelady et al. (2009) *(19)*	unclear	unclear	high	unclear	low	unclear	low	**unclear**	⊕⊕

GRADE score and interpretation
⊕⊕⊕⊕ **High.** We are very confident that the true effect lies close to that of the estimate of the effect.
⊕⊕⊕O **Moderate.** We are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
⊕⊕OO **Low.** Our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
⊕OOO **Very low.** We have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

Table 3. Risk of bias and GRADE assessments.

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Study quality for the various outcomes tested was generally low or very low by GRADE score (Table 3), meaning we had low or very low confidence in the effect estimates for all outcomes.

Unsupervised and supervised studies 6/9 unsupervised intervention studies (17,22,24,25,27,32) and 5/16 supervised intervention studies (19,20,28,37,38,14), showed an effect of the PA intervention tested on their specified primary outcome (Table 4.) LTFU was higher (14.5% vs 10%) and adherence to the intervention was lower (73.6% vs. 86%) for unsupervised vs. supervised studies.

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Study	Supervised/ monitored/unsupervised	LTFU intervention group(s)	Adherence to intervention	Exercise effect on primary outcome? (by CI or p<0.05)
Unsupervised studies				
Tripette et al. (2014) *(17)*	Unsupervised	Not reported	Not reported	**Yes (p<0.001)**
O'Toole et al. (2003) *(32)*	Unsupervised	4/21 (19%)	17/21 (81%)	**Yes (p<0.05)**
Lewis et al. (2014) *(26)*	Unsupervised	5/66 (8%)	Not reported	**No**
Daley et al. (2015) *(27)*	Unsupervised	4/47(8.5%)	41/47 (87%)	**Yes (p=0.03)**
Youngwanichsetha et al. (2013) *(22)*	Unsupervised	2/32 (6%)	Not reported	**Yes (p=0.02)**
Ashrafinia et al. (2014) *(24)*	Unsupervised	Not reported	Not reported	**Yes (p<0.001)**
McIntyre et al. (2012) *(31)*	Unsupervised	1/15 (7%)	6/14 (40%)	**No**
Ashrafini et al. (2015) *(25)*	Unsupervised	Not reported	Not given	**Yes (p<0.001)**
Lovelady et al. (2009) *(21)*	Unsupervised	3/10 (30%)	83.4% aerobic sessions	**No**
		14.5% (reported by 6/9)	**73.6% (reported by 4/9)**	**6/9 showed effect**
Supervised studies				
Brekke et al. (2014) *(18)*	HR monitor	1/34 (3%)	62/68 (91%)	**No**
Bertz et al. (2012) *(35)*	HR monitor	2/34 (6%)	Not reported	**No**
Da Costa et al. (2009) *(29)*	HR monitor	6/46 (13%)	35/46 (76%)	**No**
Dritsa et al. (2008) *(19)*	HR monitor	6/46 (13%)	Not reported	**Yes (p0.016)**
McCrory et al. (1999) *(36)*	HR monitor	2/44 (5%)	21/22 (95%)	**No**
Daley et al. (2008) *(30)*	Pedometer	4/20 (20%)	Not reported	**No**
Kinnunen et al. (2007) *(28)*	Unsupervised, optional supervised	5/53 (9%)	Not reported	**No**
Heh et al. (2008) *(23)*	Supervised and unsupervised	2/35 (5.7%)	Not reported	**Yes (p=0.01)**

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Osman et al. (2016) *(15)*	Supervised	Not reported	Not reported	**No**
Osman et al. (2020) (14)	Supervised	Not reported	Not reported	**No**
Armstrong et al. (2003) *(20)*	Supervised, Pedometer	Not reported	64% attendance	**Yes (p<0.01)**
Norman et al. (2010) *(33)*	Supervised	2/62 (3%)	130/130 (100%)	**No**
Zourladani et al. (2012) *(37)*	Supervised	Not reported	Not reported	**Yes (p<0.001)**
Lovelady et al. (1995) *(38)*	Supervised	Not reported	Not reported	**Yes (p<0.0001)**
Dewey et al. (1994) *(39)*	Supervised	Not reported	149/165 sessions completed (90%)	**No**
Lovelady et al. (2000) *(34)*	Supervised	6/27 (22%)	Not given	**No**
		9.97% (reported by 10/16)	**86% (reported by 6/16)**	**5/16 showed effect**

Table 4. Comparison of studies using unsupervised and supervised interventions for mean % lost to follow-up (LTFU), mean % adherence and effect of interventions on primary outcome.

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Discussion

In this systematic review, we show that PA interventions have inconsistent effects on most outcomes, with generally low study quality and high risk of bias. Adherence was reported by only 9/25 studies, and LTFU for 16/25 studies. For any intervention, the LTFU and adherence must be reported to allow judgement on whether the intervention was acceptable to participants and whether there can be confidence in the effect, or lack of it, shown.

Adjustment for known confounders, such as pre-existing exercise habits, physical fitness, weight, BMI were either not present or unclear in the majority of studies. This could have contributed to the heterogeneity of study conclusions.

Supervised exercise interventions did not appear to be more likely to show effect on the various outcomes tested than unsupervised interventions. Studies evaluating unsupervised interventions had lower LTFU and higher adherence, which increases our confidence in the outcomes of the studies. This was unexpected and may reflect that women may feel more empowered when allowed to exercise on their own terms; self-efficacy may lead to greater efficacy of the intervention.

Agreement for effect on primary outcomes in ≥2 studies was evident for PA interventions improving physical fitness (37,38) and reducing fatigue (19,25). Single studies were identified which reported that PA improved glycaemia (22) and sleep quality (25). These findings are consistent with changes in physiology which would be expected with increased physical activity and as seen in other populations(40,41).

Three studies showed no effect of physical activity interventions on breast milk production, intake and cortisol levels (14,36,39) which is encouraging for women who wish to undertake recommended physical activity levels in the postnatal period and breastfeed.

Women considering postnatal physical activity can be counselled that physical activity is unlikely to have an impact on breastfeeding or infant growth. Additionally, no studies led to any serious adverse events. Whilst the wide range of interventions makes recommendations on the types of exercise difficult to assess, it also demonstrates the inventiveness of researchers in the field who have considered using video games (17), pilates (24,25), tai chi (22) and telephone-based interventions (26) in addition to more conventional exercise counselling and supervised exercise classes.

Since the majority of these studies were conducted, there are now many wearable devices available, which could improve objective assessment of PA in intervention studies. These will allow accurate assessment of adherence and measures of fitness to be measured, such as morning resting heart rate. Our group is conducting feasibility studies before, during and after pregnancy using wearables and phone applications to monitor cardiovascular health.

Strengths of this study included a published protocol and a literature search which was updated close to publication. In order to provide information relevant to clinicians and policy makers looking to improve physical activity in postnatal women, we focussed our study on moderate-vigorous exercise programs which were evaluated for effect on a range of mental and physical health parameters. A key factor in designing and costing physical activity interventions is whether they are supervised or unsupervised, our study allows these groups to be compared. The weakness of this study is that the studies included were generally small in size and of largely low quality and high or uncertain risk of bias. This review considered only PA intervention studies which were compared to a control group with no PA intervention or with advice to be sedentary, to allow evaluation of effect of the PA intervention on the primary outcomes of each study. Our inclusion of only women in the postnatal

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year meant that we did not included studies which compares PA interventions in groups other than postnatal women which had high adherence and positive or negative effects and will have been excluded from our analysis. Inclusion od only PA interventions in the year after delivery may also have potentially excluded studies of interventions later after delivery which may be appealing to women and effective in improving physical and mental health.

Further research is needed on the optimal timing, method of delivery and content of postnatal PA interventions to maximise recruitment, adherence and ensure effects on outcomes are effectively evaluated. Weight gain is encouraged in pregnancy, but complications such as excessive gestational weight gain, preeclampsia, or gestational diabetes may help identify women likely to benefit from PA interventions to improve their long-term health. Studies must report using internationally agreed standards and outcome sets to improve their utility to women and their healthcare professionals.

More generally, the postnatal period is a difficult time restart and maintain physical activity. Public health approaches which encourage and enable women to increase their PA after having a baby could have broader population impacts with benefits seen in short- and long-term risk of chronic diseases; However, studies to date have not consistently demonstrated these effects. None of the included studies referred to Public and Patient Involvement (PPI) in the design or conduct of their research. PPI is invaluable to ensure recruitment, adherence and rigorous evaluation with outcomes which are relevant to women and their healthcare providers.

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Competing Interests

The authors declare that that they have read and understood the BMJ Group policy on declarations of interest and have no competing interests to declare.

Ethical approval

Ethical approval was not required for this review article.

Contributorship statement

Edward Mullins, Shalini Sharma and Alison MacGregor were involved in study design and conduct, data analysis, report drafting and editing. EM and SS conducted the literature review.

Data Sharing

Not applicable for this narrative review

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Dissemination We will disseminate the information in this study to women in our area via our institutional news and social media channels.

Data sharing We will share our data on request to the corresponding author.

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1. postpartum period/ or lactation/
2. postpartum period.mp.
3. Peripartum Period/
4. peripartum period.mp.
5. postpartum women.mp.
6. post partum women.mp.
7. (post natal or postnatal).tw.
8. (postpartum or post partum).tw.
9. (post pregnancy or postpregnancy).tw.
10. ((Post or after or follow*) adj birth*).tw.
11. ((Post or after or follow*) adj labo?r).tw.
12. ((Post or after or follow*) adj childbirth*).tw.
13. ((Post or after or follow*) adj pregnancy).tw.
14. ((Post or after or follow*) adj parturition).tw.
15. ((Post or after or follow*) adj delivery).tw.
16. (interpregnanc* or inter pregnanc* or (between adj pregnanc*)).tw.
17. (interconception or inter conception or (between adj conception)).tw.
18. puerperium.tw.
19. or/1-18
20. exercise/ or cool-down exercise/ or gymnastics/ or muscle stretching exercises/ or physical conditioning, human/ or circuit-based exercise/ or high-intensity interval training/ or plyometric exercise/ or resistance training/ or running/ or jogging/ or swimming/ or walking/ or warm-up exercise/ or pronation/ or supination/
21. exercise.tw

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22. exercise therapy/ or motion therapy, continuous passive/

23. motor activity/ or physical exertion/ or physical fitness/ or cardiorespiratory fitness/ or postural balance/

24. sports/ or athletic performance/ or baseball/ or basketball/ or bicycling/ or boxing/ or football/ or golf/ or gymnastics/ or hockey/ or martial arts/ or tai ji/ or mountaineering/ or racquet sports/ or tennis/ or return to sport/ or running/ or jogging/ or skating/ or snow sports/ or skiing/ or soccer/ or sports for persons with disabilities/ or "track and field"/ or volleyball/ or walking/ or water sports/ or swimming/ or diving/ or weight lifting/ or wrestling/ or youth sports/

25. sports medicine/ or sports nutritional sciences/

26. fitness.tw.

27. (activit* and program*).tw.

28. (locomotor and activit*).tw.

29. (motor and activit*).tw.

30. (structure* and activit*).tw.

31. training*.tw.

32. (therap* and exercis*).tw.

33. physical*.tw.

34. (activit* or conditioning* or effort* or exert*).tw.

35. 33 and 34

36. 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 35

37. randomized controlled trial'.pt.

38. (random$ or placebo$ or single blind$ or double blind$ or triple blind$).mp.

39. (retraction of publication or retracted publication or controlled clinical trial).pt. or randomi?ed.tw. or randomly.tw.

40. controlled clinical trial.pt.

41. randomi?ed.tw.

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42. randomly.tw.

43. or/37-42

44. exp animals/ not humans.sh.

45. (random sampl$ or random digit$ or random effect$ or random survey or random regression).mp. not 'randomized controlled trial'.pt.

46. 43 not (44 or 45)

47. 19 and 36 and 46

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Embase search strategy

1. postnatal care/ or lactation/
2. postnatal care.mp.
3. perinatal period/
4. perinatal period.mp.
5. postpartum period.mp.
6. peripartum period.mp.
7. postpartum women.mp.
8. post partum women.mp.
9. (postnatal or post natal).ti,ab.
10. (post pregnancy or postpregnancy).ti,ab.
11. ((Post or after or follow*) adj birth*).ti,ab.
12. ((Post or after or follow*) adj labo?r).ti,ab.
13. ((Post or after or follow*) adj childbirth*).ti,ab.
14. ((Post or after or follow*) adj pregnancy).ti,ab.
15. ((Post or after or follow*) adj parturition).ti,ab.
16. ((Post or after or follow*) adj delivery).ti,ab.
17. (interpregnanc* or inter pregnanc* or (between adj pregnanc*)).ti,ab.
18. (interconception or inter conception or (between adj conception)).ti,ab.
19. puerperium/
20. puerperium.ti,ab.
21. or/1-20
22. exercise/ or aerobic exercise/ or anaerobic exercise/ or aquatic exercise/ or arm exercise/ or athletic performance/ or breathing exercise/ or circuit training/ or closed kinetic chain exercise/ or cool down/ or dynamic exercise/ or endurance

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training/ or high intensity interval training/ or isokinetic exercise/ or leg exercise/ or muscle exercise/ or open kinetic chain exercise/ or pilates/ or plyometrics/ or resistance training/ or static exercise/ or warm up/

23. exercise.ti,ab.

24. kinesiotherapy/ or pelvic floor muscle training/ or stretching exercise/

25. athletic performance/

26. sport/ or aquatic sport/ or athletics/ or baseball/ or basketball/ or body building/ or combat sport/ or contact sport/ or "cricket (sport)"/ or cycling/ or diving/ or endurance sport/ or extreme sport/ or football/ or golf/ or hockey/ or horseback riding/ or ice hockey/ or jogging/ or motor sport/ or racquet sport/ or rock climbing/ or roller skating/ or rowing/ or rugby/ or running/ or skateboarding/ or skating/ or skiing/ or soccer/ or swimming/ or tai chi/ or team sport/ or triathlon/ or volleyball/ or water skiing/ or winter sport/ or yoga/ or youth sport/

27. sports medicine/

28. fitness/

29. fitness.ti,ab.

30. (activit* and program*).ti,ab.

31. (locomotor and activit*).ti,ab.

32. (motor and activit*).ti,ab.

33. (structure* and activit*).ti,ab.

34. training*.ti,ab.

35. (therap* and exercis*).ti,ab.

36. physical*.ti,ab.

37. (activit* or conditioning* or effort* or exert*).ti,ab.

38. 36 and 37

39. 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 38

40. (random$ or placebo$ or single blind$ or double blind$ or triple blind$).ti,ab

41. RETRACTED ARTICLE/

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42. randomi?ed.ti,ab.

43. randomly.ti,ab.

44. or/40-43

45. exp animal$/ not human$.sh,hw.

46. (random sampl$ or random digit$ or random effect$ or random survey or random regression).ti,ab. not exp randomized controlled trial/

47. 44 not (45 or 46)

48. 21 and 39 and 47

49. limit 48 to embase

50. limit 49 to exclude medline journals

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CENTRAL SEARCH STRATEGY

1. MeSH descriptor: [Postpartum Period] explode all trees
2. MeSH descriptor: [Lactation] explode all trees
3. (postpartum period)
4. MeSH descriptor: [Peripartum Period] explode all trees
5. (peripartum period)
6. (postpartum women)
7. (post partum women)
8. ((post natal or postnatal)):ti,ab,kw
9. ((post partum or postpartum)):ti,ab,kw
10. ((post pregnancy or postpregnancy)):ti,ab,kw
11. (((Post or after or follow*) NEXT birth*)):ti,ab,kw
12. (((Post or after or follow*) NEXT labo?r)):ti,ab,kw
13. (((Post or after or follow*) NEXT childbirth*)):ti,ab,kw
14. (((Post or after or follow*) NEXT pregnancy)):ti,ab,kw
15. (((Post or after or follow*) NEXT parturition)):ti,ab,kw
16. (((Post or after or follow*) NEXT delivery)):ti,ab,kw
17. ((interpregnanc* or inter pregnanc* or (between NEXT pregnanc*))):ti,ab,kw
18. ((interconception or inter conception or (between NEXT conception))):ti,ab,kw
19. (puerperium):ti,ab,kw
20. #1 or #2 or #3 or #4 or #5 #8 or #9 or #10 or #11 or #12 or #13 or #14 or #15 or #16 or #17 or #18 or #19
21. MeSH descriptor: [Exercise] explode all trees
22. (exercise):ti,ab,kw

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23. MeSH descriptor: [Exercise Therapy] explode all trees

24. MeSH descriptor: [Physical Fitness] explode all trees

25. MeSH descriptor: [Sports] explode all trees

26. MeSH descriptor: [Sports Medicine] this term only

27. MeSH descriptor: [Sports Nutritional Sciences] this term only

28. (fitness):ti,ab,kw

29. ((activit* and program*)):ti,ab,kw

30. ((locomotor and activit*)):ti,ab,kw

31. ((motor and activit*)):ti,ab,kw

32. ((structure* and activit*)):ti,ab,kw

33. (training*):ti,ab,kw

34. ((therap* and exercis*)):ti,ab,kw

35. (physical*):ti,ab,kw

36. ((activit* or conditioning* or effort* or exert*)):ti,ab,kw

37. #35 and #36

38. #21 or #22 or #23 or #24 or #25 or #26 or #27 or #28 or #29 or #30 or #31 or #32 or #33 or #34 or #37

39. #20 and #38

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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	yes
Synthesis of results	6	Specify the methods used to present and synthesise results.	yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	yes
Interpretation	10	Provide a general interpretation of the results and important implications.	yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	Yes
Registration	12	Provide the register name and registration number.	Yes

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

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Synthesis Without Meta-analysis (SWiM) reporting items

The citation for the Synthesis Without Meta-analysis explanation and elaboration article is: Campbell M, McKenzie JE, Sowden A, Katikireddi SV, Brennan SE, Ellis S, Hartmann-Boyce J, Ryan R, Shepperd S, Thomas J, Welch V, Thomson H. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline BMJ 2020;368:l6890 http://dx.doi.org/10.1136/bmj.l6890

SWiM is intended to complement and be used as an extension to PRISMA			
SWiM reporting item	**Item description**	**Page in manuscript where item is reported**	**Other***
Methods			
1 Grouping studies for synthesis	1a) Provide a description of, and rationale for, the groups used in the synthesis (e.g., groupings of populations, interventions, outcomes, study design)	P4	
	1b) Detail and provide rationale for any changes made subsequent to the protocol in the groups used in the synthesis	n/a	
2 Describe the standardised metric and transformation methods used	Describe the standardised metric for each outcome. Explain why the metric(s) was chosen, and describe any methods used to transform the intervention effects, as reported in the study, to the standardised metric, citing any methodological guidance consulted	P4	
3 Describe the synthesis methods	Describe and justify the methods used to synthesise the effects for each outcome when it was not possible to undertake a meta-analysis of effect estimates	P4	
4 Criteria used to prioritise results for summary and synthesis	Where applicable, provide the criteria used, with supporting justification, to select the particular studies, or a particular study, for the main synthesis or to draw conclusions from the synthesis (e.g., based on study design, risk of bias assessments, directness in relation to the review question)	n/a	

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Synthesis Without Meta-analysis (SWiM) reporting items

SWiM reporting item	Item description	Page in manuscript where item is reported	Other*
5 Investigation of heterogeneity in reported effects	State the method(s) used to examine heterogeneity in reported effects when it was not possible to undertake a meta-analysis of effect estimates and its extensions to investigate heterogeneity	P4	
6 Certainty of evidence	Describe the methods used to assess certainty of the synthesis findings	n/a	
7 Data presentation methods	Describe the graphical and tabular methods used to present the effects (e.g., tables, forest plots, harvest plots). Specify key study characteristics (e.g., study design, risk of bias) used to order the studies, in the text and any tables or graphs, clearly referencing the studies included	Tables	
Results			
8 Reporting results	For each comparison and outcome, provide a description of the synthesised findings, and the certainty of the findings. Describe the result in language that is consistent with the question the synthesis addresses, and indicate which studies contribute to the synthesis	P8	
Discussion			
9 Limitations of the synthesis	Report the limitations of the synthesis methods used and/or the groupings used in the synthesis, and how these affect the conclusions that can be drawn in relation to the original review question	P17	

PRISMA=Preferred Reporting Items for Systematic Reviews and Meta-Analyses.
*If the information is not provided in the systematic review, give details of where this information is available (e.g., protocol, other published papers (provide citation details), or website (provide the URL)).

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